Dr.-Ing. Juan Alberto Cabrera Guerrero

Curriculum Vitae

Current position	Junior Research Group Leader in CeTI since 10/2026
Current institution	Centre for Tactile Internet with Human-in-the-Loop (CeTI) of Technische Universität Dresden, Germany
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Qualifications and Career

Qualifications

2022	**Dr.-Ing.**, Communication Networks, Technische Universität Dresden, Germany. Date: 16.06.2022. Supervisor: Professor Dr.-Ing. Dr. h.c. Frank H.P. Fitzek. Subject: On the Design of Future Communication Systems with Coded Transport, Storage, and Computing.
2015	**M.Sc.**, Wireless Communication Systems, Aalborg University, Denmark.
2013	**Electrical Engineer**, focused on communication theory, Universidad Simón Bolívar, Venezuela.

Stages of academic career

10/2026 – today	**Junior Research Group Leader**, Centre for Tactile Internet with Human-in-the-Loop (CeTI), Technische Universität Dresden, Germany.
2023 – today	**Managing Director**, Ecologic-Computing GmbH. Spin-off startup focusing on reducing energy consumption in communication systems.
2022 – today	**Postdoc**, leading the post-Shannon Communication and Analogue Computing Research Group at the Deutsche Telekom Chair of Communication Networks at Technische Universität Dresden, Germany.
2015 – 2022	**Research Assistant**, Technische Universität Dresden, Germany.

Teaching and Supervision

2022–2026	As a Postdoc, I helped in the guidance and supervision of 5 Doctoral Researchers
2016–2026	Supervision of 15+ M.Sc. theses and Seminar work
SS 2027–	Course responsible: Advanced Communication Networks (M.Sc.) at TU Dresden, taking over the lectures and hands-on exercises on network function virtualization, software-defined networking, edge computing, network slicing, time-sensitive networking, and 5G
2017–today	Co-creator of the M.Sc. course Future Communication Networks (formerly Communication Networks 3) at TU Dresden, with a curriculum built from ongoing IETF standardization work. Since 2022, lectures on post-Shannon communication and analog computing
2016–2025	Teaching Course: Practical Implementations of Network Coding at TU Dresden
2018	Teaching Course: Advanced Topics in Signal Processing at TU Munich

Network architectures, softwarization, and experimental platforms. These two papers show how I approach networks: as programmable infrastructure in which communication and computing are designed together. In the first, I describe how software-defined networking, network function virtualization, and network coding can be combined in the mobile edge cloud to meet the latency and resilience requirements of the Tactile Internet. The second presents ComNetsEmu, an open-source emulator for software-defined and virtualized networks that runs on commodity hardware. I helped build and evaluate it, and I have taught its use in several IEEE tutorials together with Frank H. P. Fitzek and Fabrizio Granelli.

[1] J. A. Cabrera, R.-S. Schmoll, G. T. Nguyen, et al. "Softwarization and Network Coding in the Mobile Edge Cloud for the Tactile Internet". In: *Proc. IEEE* 107.2 (Feb. 2019), pp. 350–363. DOI: 10.1109/JPROC.2018.2869320

[2] Z. Xiang, S. Pandi, J. A. Cabrera, et al. "An Open Source Testbed for Virtualized Communication Networks". In: *IEEE Commun. Mag.* 59.2 (Feb. 2021), pp. 77–83. DOI: 10.1109/MCOM.001.2000578

Resilient transport and storage. Storage is communication across time, so the codes that make transport reliable can also keep stored information available. OneCode uses a single random linear network code to generate packets, recode them over several hops, and repair the redundancy that is lost when nodes fail. It results from our DFG SPP 2378 project "One Code to Rule Them All," and its evaluation rests on analysis and simulation. The second paper studies peers at the network edge that cache network-coded content and leave or fail over time. We define a measure, the grade, of how far the coded cache has degraded, and we use it to decide when to refresh the cache. I worked closely with the two doctoral researchers who led this line of work.

[3] J. A. Cabrera, I. Yilmaz, and F. H. P. Fitzek. "OneCode Architecture: A Single RLNC Regenerating Code for Storage and Transport". In: *Proc. Eur. Wireless Conf. (EW)*. Sophia Antipolis, France, Oct. 2025

[4] S. Zimmermann, P. Schwenteck, J. A. Cabrera, et al. "Grade to the Edge: How Many Unreliable Nodes Does It Take to Break a Content Delivery Network?" In: *Proc. IEEE Int. Symp. World Wireless, Mobile Multimedia Netw. (WoWMoM)*. Belfast, U.K., June 2022, pp. 331–336. DOI: 10.1109/WoWMoM54355.2022.00068

Task-oriented communication with identification codes. Does a receiver need the whole message to perform its task? In message identification, it only checks whether one event of interest occurred, which requires far fewer bits than transmitting the message. The first paper proposes performance metrics for identification codes that are closer to practical systems than the asymptotic bounds. The second is a tutorial that opens these codes to communication engineers outside information theory. I gave day-to-day scientific guidance to the doctoral researcher who led both. The third paper, joint work with the Chair of Communication Networks at TU Munich, applies identification codes to a truck platoon in a cellular vehicle-to-infrastructure setting. The vehicles send short tags to verify their local control decisions against an edge server at the base station. In simulation, this hybrid scheme needed about four times less traffic than fully centralized control.

[5] C. v. Lengerke, A. Hefele, J. A. Cabrera, et al. "Beyond the Bound: A New Performance Perspective for Identification via Channels". In: *IEEE J. Sel. Areas Commun.* 41.8 (June 2023), pp. 2687–2706. DOI: 10.1109/JSAC.2023.3288239

[6] C. von Lengerke, J. A. Cabrera, M. Reisslein, et al. "Codes for Identification via Channels: Tutorial for Communications Generalists". In: *IEEE Commun. Surveys Tuts.* 28 (2026), pp. 181–223. DOI: 10.1109/COMST.2025.3579815

[7] P. Kutsevol, C. von Lengerke, J. A. Cabrera, et al. "Distributed Platoon Control via Semantic-Aware Identification Codes". In: *Proc. IEEE Int. Conf. Commun. (ICC)*. Montreal, Canada, June 2025. DOI: 10.1109/ICC52391.2025.11161806

Functional and semantic compression. I proposed functional compression for networked applications as a research direction and guided the doctoral work that developed it. The first paper merges the views of several cameras at edge application servers through multilevel semantic compression and maps the design onto the 3GPP architecture for edge computing in 5G, in collaboration with Nokia. Its evaluation uses an open-source prototype on commodity hardware with a simulated network. The second is a demonstration of functional compression in the control loop of a real balancing robot, which received the Best Demo Award at IEEE CCNC 2024.

[8] S. Rezwan, H. Wu, J. A. Cabrera, et al. "MLcXR+: Multilevel Semantic Compression for 3D Immersion Over 5G Networks". In: *IEEE Access* 13 (2025), pp. 164771–164786. DOI: 10.1109/ACCESS.2025.3610029

[9] S. Rezwan, J. A. Cabrera, and F. H. P. Fitzek. "Balancing Beyond-Shannon: Demonstration of Functional Compression Using a Balancing Robot". In: *Proc. IEEE Consum. Commun. Netw. Conf. (CCNC)*. Best Demo Award. Las Vegas, USA, Jan. 2024, p. 2. DOI: 10.1109/CCNC51664.2024.10454858

Function computation over networks. Can nodes test whether they agree without exchanging their full states? Together with Holger Boche's group at TU Munich, we studied this question for relay networks. I contributed the systems perspective and co-developed the link between message identification and the computation of a function chosen by the receiver. The mathematical modeling and the proofs were led by Johannes Rosenberger.

[10] J. Rosenberger, H. Boche, J. A. Cabrera, et al. "Stochastic Consensus-Testing in Relay Networks". In: *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*. 2025. DOI: 10.1109/ISIT63088.2025.11195504

Patents and Research Data

Technology transfer through patents. The first two results document the transfer of my research on task-oriented communication and computation-aware networking towards protected technical inventions. The granted German patent concerns a computer selector that chooses an appropriate computing substrate, such as analogue, digital, biological, or other unconventional computing platforms, according to task-specific requirements including latency, energy consumption, reliability, trustworthiness, and implementation constraints. The second result is a published international patent application on multi-agent consensus testing using message identification, extending my research beyond-Shannon communication towards distributed decision-making systems.

[1] J. A. Cabrera, P. Hofmann, F. H. P. Fitzek, et al. "Calculation System and Related Procedure". German patent DE102023114253B3. Technische Universität Dresden and Technische Universität München. Aug. 2024

[2] J. Rosenberger, J. A. Cabrera, F. H. P. Fitzek, et al. "Communication Devices and Methods for Message Identification". International patent application WO 2026/114806 A1. Technische Universität Dresden and Technische Universität München. June 4, 2026

FAIR research data and reproducibility. The following two results are published datasets from my research team. They show my commitment to FAIR data handling, reproducibility, and open scientific infrastructure. Both datasets provide measurement data from software-defined-radio and channel-emulation experiments.

[3] P. K. H. Sheshagiri, M. Reisslein, J. A. Cabrera, et al. *Raw channel metrics from Ettus N210 USRP for common randomness generation*. Dec. 19, 2024. DOI: 10.21227/tn1p-p947. published

[4] P. K. H. Sheshagiri, M. Reisslein, J. A. Cabrera, et al. *RSS readings in Software Defined Radios with Channel Emulator for CR Generation*. Jan. 1, 2026. DOI: 10.21227/vmdc-dw42. published

Academic Distinctions

2024	Best Demo Award at the IEEE Consumer Communications & Networking Conference (CCNC)
2022	Awarded the Funding for the BMBF Project "Software Campus," 2021, for a project on coded distributed computing
2022	Best Paper of the First International Workshop on Technologies for Network Twins (TNT), co-located with IEEE/IFIP NOMS 2022 Conference
currently	h-index: 18, i10-index: 29, number of citations: 1100+, 29 journal publications.

Technology Transfer

Co-founder and Managing Director of Ecologic-Computing GmbH. The company received BMFTR StartUpConnect research-transfer funding for PoShCodes, "Specialised Post-Shannon Codes for Highly Efficient Goal-Oriented Communication" (07/2024–06/2026, €0.67 million). The project develops identification-theoretic algorithms and protocols to reduce the amount of data that must be transmitted in communication systems and to transfer these principles towards hardware implementations. My role includes scientific leadership, project coordination, and research-to-application strategy.